System Validation Report

1. Purpose

System validation shows that the finished Online Bookstore meets the requirements in the SRS. This report joins the Requirements Traceability Matrix (docs/RTM.md) to the system-level evidence: the end-to-end journey test, the security tests, the performance test and the user acceptance test.

2. System-level test results

Testing activity	Result
Full Django test suite	260 passed, 1 skipped (261 total)
End-to-end test	1 passed
Security tests	16 passed
Performance test	50 users, 30 seconds, 13,040 requests, 0 errors

The end-to-end test registers a user, verifies the email, logs in, searches, fills the cart, adds an address, checks out, has an admin ship the order, and has the buyer write a review that an admin then hides. The security tests cover access control, odd and malicious input, tampered tokens, CSRF, password hashing and card-data protection.

3. Performance validation (BOOK-NF-001, BOOK-NF-004)

Target: 90th percentile response time of 2,000 ms or less. Latest run: 50 users for 30 seconds, 13,040 requests, 0 errors. Earlier recorded runs from tests/perf/results.md are shown for comparison.

Run	Result recorded by load_test.py
2026-10-30 18:27 - 1 users for 10s	72 requests in 10s (7.1 requests/second), 0 errors. Overall 90th percentile: 54 ms (target 2000 ms). Result: PASS
2026-10-30 19:05 - 50 users for 30s	7831 requests in 30s (259.6 requests/second), 0 errors. Overall 90th percentile: 142 ms (target 2000 ms). Result: PASS
2026-10-30 19:07 - 200 users for 30s	7392 requests in 31s (235.6 requests/second), 0 errors. Overall 90th percentile: 1636 ms (target 2000 ms). Result: PASS

The first runs on Django's development server failed the target; switching to gunicorn with WhiteNoise fixed it. The 1,000-concurrent-user requirement (BOOK-NF-004) was not demonstrated on a laptop; 200 users were tested.

4. User acceptance test

Total scenarios	Passed	Failed	Not run
26	26	0	0

Scenarios U-01 to U-26 are defined in docs/UAT.md and cover authentication, catalog and search, cart and checkout, orders, payments, seller and admin roles, reviews, security, keyboard use and zoom.

5. Requirement validation (from the RTM)

Req ID	Requirement	Verified by	Status	Comment
BOOK-F-001	Register with email verification	accounts.test_registration, bookstore.test_e2e	**Met**	The email is sent through Celery. See DEVIATIONS D8 and K1.
BOOK-F-002	Authenticate with hashed password and JWT	accounts.test_token_auth, accounts.tests	**Met**	Session login for the pages, JWT for API clients. See D1.
BOOK-F-003	Lock the account after 5 failed logins for 15 minutes	accounts.tests (lockout), accounts.test_token_auth	**Met**	Both logins share one implementation.
BOOK-F-010	Paginated catalog, 20 per page	catalog.test_api, catalog.tests, catalog.test_detail, catalog.test_caching, catalog.test_cache_invalidation, storefront.tests	**Met**	Load test results are under BOOK-NF-001.
BOOK-F-011	Search by title, author, ISBN or genre, with filter and sort	catalog.test_search, bookstore.test_security (fuzzing)	**Met**	Sort by price, newest, title and rating.
BOOK-F-020 to 023	Cart and wishlist, cart kept across sessions	cart.test_cart, cart.tests, wishlist.tests, storefront.test_wishlist_page, bookstore.test_e2e	**Met**	

Req ID	Requirement	Verified by	Status	Comment
BOOK-F-030 to 035	Checkout, coupons, payment, confirmation and invoice	orders.tests, notifications.test_sms, bookstore.test_e2e	**Met**	Razorpay test mode (D6). The invoice is an API endpoint.
BOOK-F-040 to 044	Order history, tracking, returns and refunds, fulfilment	orders.tests, orders.test_fulfilment	**Met**	Fulfilment is per order (D12). History and returns are API only.
BOOK-F-050 to 054	Seller and admin dashboards, analytics	accounts.test_admin, catalog.test_seller_admin, reviews.test_admin, orders.test_fulfilment	**Partly met**	Sales analytics exists only as a prototype in seller-analytics/ (D11).
BOOK-F-060 to 062	Ratings, reviews and moderation	reviews.test_reviews, reviews.test_moderation, reviews.tests, storefront.test_book_detail	**Partly met**	Moderation works. The "verified buyers only" rule is not enforced (D13).
BOOK-NF-001	Catalog and product pages in 2 s or less for 90% of requests	tests/perf/load_test.py (results in tests/perf/results.md), catalog.test_caching	**Met**	90th percentile 142 ms at 50 users and 1,636 ms at 200 users.
BOOK-NF-002	99.5% availability per month	monitoring.tests	**Partly met**	Availability can only be measured in production.
BOOK-NF-003	PCI-DSS: no card data in plaintext on our servers	bookstore.test_security.DataProtectionTests	**Met**	A formal PCI-DSS audit applies only to a production deployment.
BOOK-NF-004	At least 1,000 concurrent users	tests/perf/load_test.py	**Partly met**	200 users tested. 1,000 users cannot be shown on a laptop.
BOOK-NF-005	WCAG 2.1 AA on browse, cart and checkout	docs/accessibility/WCAG-2.1-AA-report.md, UAT scenarios U-23 and U-24	**Partly met**	Becomes A when the manual keyboard and zoom checks in the UAT script are recorded as passed.
BOOK-SR-001	TLS 1.2+ for all connections	bookstore.test_security.AccessControlTests (includes CSRF)	**Partly met**	The TLS version is enforced by the web server or host. See D4.
BOOK-SR-002	Salted bcrypt or argon2 password hashes	accounts.test_token_auth.PasswordHashingTests, bookstore.test_security.DataProtectionTests	**Met**	
BOOK-SR-003	Card data never stored	bookstore.test_security.DataProtectionTests	**Met**	The test scans every model for card-related field names.
BOOK-SR-004	Role-based access control	accounts.test_rbac, accounts.test_rbac_endpoints, bookstore.test_security.AccessControlTests, catalog.test_seller_admin	**Met**	
BOOK-SR-005	Admin actions logged with time, user and action, kept for 1 year	accounts.test_audit, accounts.test_admin	**Met**	

14 requirement groups met, 6 partly met, 0 not met.

6. Known limitations

- Sales analytics exists as a prototype in seller-analytics/, not inside the application.
- The "verified buyers only" rule for reviews.
- Order history and return requests are API endpoints without storefront pages.
- End-to-end tests run without a real browser. A browser smoke test (Playwright) is a possible next step.
- A production deployment, a TLS certificate, real uptime monitoring and the 1,000-user load test.

7. Conclusion

All system-level tests passed. Fourteen of twenty requirement groups are fully met and six are partly met, for the reasons given in section 5; none is unmet.